

MISSION: EARTH

Software Game Suite - Product & Technical Design Specification

Three independent, fast-play software games designed to attract students, introduce GRSS concepts, and keep booth throughput high.

GRSS OPEN DAY / GAME SELECT

PIXEL HUNTER

EARTH UNDER WATCH

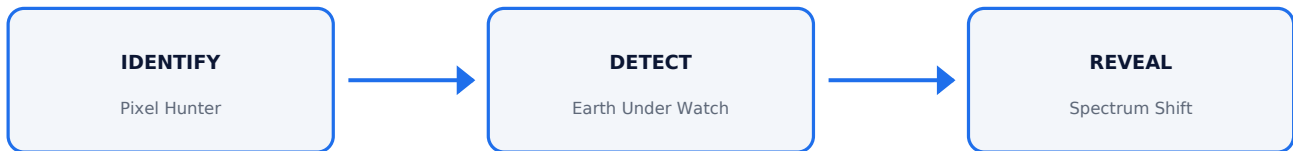
SPECTRUM SHIFT

Audience	Students of mixed technical backgrounds; no prior GRSS knowledge assumed.
Format	Browser-based, fullscreen software games; mouse / keyboard / touch friendly.
Target session	30-90 seconds per game; all three can be experienced in about 3-4 minutes.
Product principle	Instantly understandable gameplay, short feedback loops, strong visual identity, genuine GRSS relevance.
Positioning	Explicitly different from GeoQuest: no auction, no multi-round strategy structure, no long rules briefing.

Prepared as the team handoff specification for implementation, content preparation, testing, and Open Day operations.

01 Product North Star

The GRSS Open Day booth should feel like a small interactive lab rather than a quiz counter. The student experience is designed around three different verbs: identify, detect, and reveal. Each game teaches a real idea from geoscience and remote sensing through its mechanics instead of relying on long explanations.



Design goals

- Zero-friction onboarding: a volunteer can explain each game in one sentence.
- Fast throughput: target 30-90 seconds per attempt.
- High visual pull: a passer-by should understand that something is happening just by watching the screen.
- Replayability: randomized or rotating content, clear scores, and short rounds encourage another attempt.
- Authentic GRSS identity: imagery interpretation, change detection, spectral information, and Earth observation are embedded in the core mechanics.
- Independent development: the three games can be built by three teammates with a small shared leaderboard service.

GRSS grounding

GRSS defines its field of interest around science and engineering applied to remote sensing of the Earth, oceans, atmosphere, and space, plus processing, interpretation, and dissemination of that information. Its listed areas include land/surface processes, atmosphere and oceans, cryosphere, solid Earth/geodynamic processes, and natural disasters. The game suite deliberately samples these themes without turning the booth into a lecture.

02 Why These Three Games

Dimension	Pixel Hunter	Earth Under Watch	Spectrum Shift
Genre	Visual puzzle	Reaction / anomaly detection	Interactive spectral puzzle
Primary skill	Pattern recognition	Speed + visual attention	Experimentation + inference
Core GRSS idea	Image interpretation	Change detection / monitoring	Spectral information
Target play time	30-60 sec	45-75 sec	45-75 sec
Difficulty	Easy -> medium	Easy -> hard	Easy -> medium/hard
Best role	Entry game / crowd puller	Main attention game	Technical showcase

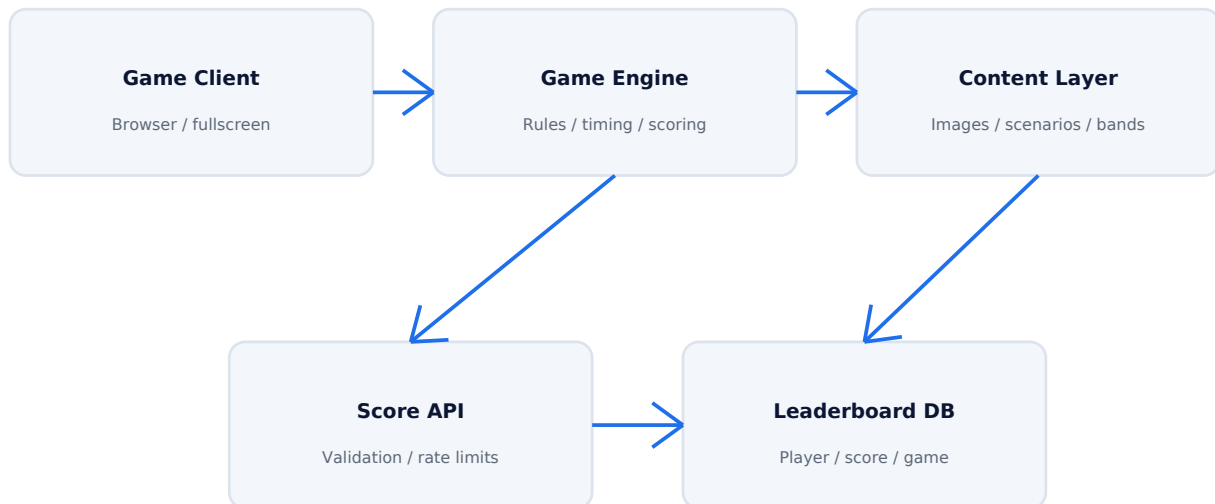
Anti-GeoQuest guardrails

- No auction mechanics, bidding, budgets, trading, or economic strategy.
- No long sequence of formal rounds or elimination stages.
- No geography trivia dependency and no requirement to know place names.
- No 5-10 minute session per player.
- No large rulebook: the UI itself teaches the mechanic.

If a student cannot understand what to do by watching one complete attempt, the game is too complicated.

03 Shared Product & Technical Architecture

The games should be separate applications at the gameplay level but share the minimum useful infrastructure: common branding, a lightweight score service, common content metadata, and an optional leaderboard. This lets three developers work in parallel without creating a monolith.



Recommended stack

- Frontend: React + TypeScript or vanilla TypeScript with a lightweight component layer. Use CSS for UI and Canvas/WebGL only where image manipulation benefits from it.
- Image processing: Canvas 2D for tiles, overlays, difference masks, and channel compositing. Keep image datasets local or preprocessed to avoid network latency at the event.
- Backend (optional but recommended): small Node.js/Express or serverless endpoint for score submission and leaderboard reads.
- Data: JSON content packs for scenarios, answers, scoring metadata, difficulty tags, and image asset paths.
- Deployment: static hosting for the games plus a small API, or fully local network deployment if campus connectivity is uncertain.
- Reliability: pre-cache game assets before opening the booth; provide a local fallback leaderboard or offline score mode.

Common UI rules

- Fullscreen-first layout with large targets and strong contrast.
- No scroll during gameplay.
- Time and score always visible but not visually dominant over the core task.
- Every click gets immediate visual feedback.
- Transitions should be 150-350 ms; avoid slow cinematic intros.
- Use one shared typography system and a consistent GRSS badge / footer.

04 GAME 01 - PIXEL HUNTER

Genre: Visual puzzle / satellite image interpretation

Tagline: How little can you see and still know?

Target duration: 30-60 seconds

Owner: Teammate 1

CORE LOOP: REVEAL -> INFER -> GUESS -> SCORE

One-line pitch

A satellite image is hidden behind tiles. The player reveals only as much as necessary to identify the Earth feature or change shown. Fewer reveals and faster guesses produce higher scores.

Player journey

PIXEL HUNTER / GAMEPLAY

Hidden satellite image grid

Reveal one tile

Confidence / score meter

Detailed rules

- Each challenge loads one image and one correct target label.
- Image starts fully obscured by a 4x4 or 5x5 tile grid.
- Player can reveal tiles in any order.
- Each reveal reduces the available score; guessing ends the challenge immediately.
- A wrong answer ends the challenge with a small penalty or zero base score; do not create long retry screens.
- After the result, auto-advance to the next challenge after 1.5-2.0 seconds.
- A complete attempt contains 3-5 images, depending on average difficulty.

Scoring model

Base: 1,000 points per image. Reveal penalty: score multiplier falls as tiles are opened. Speed bonus: add a modest time multiplier so fast pattern recognition matters. Recommended starting model:
 $\text{score} = 1000 \times \text{revealMultiplier} \times \text{speedMultiplier}$, with a floor of 100 before final answer. Tune after

live testing rather than hard-coding theoretical values.

Content taxonomy

- Land: urban expansion, agriculture, mining, deforestation, dunes, river patterns.
- Water: deltas, coastlines, flood extent, lakes, ocean features.
- Cryosphere: glaciers, snow fields, ice patterns.
- Atmosphere: cloud formations, dust events, storm signatures where visually appropriate.
- Disasters: wildfire scars, volcanic areas, flood footprints.
- Time Shift mode: show a before/after pair and ask what changed.

Difficulty ladder

- Easy: distinctive patterns and 4x4 grid.
- Medium: ambiguous natural patterns and 5x5 grid.
- Hard: false-color or time-shift image pairs.
- Expert: low-information reveal budgets and closely related answer choices.

Success criteria

At least 80% of first-time players should understand the mechanic without volunteer re-explanation; average completed attempt should stay below 60 seconds; at least 20% of players should be tempted to replay when a leaderboard is visible.

05 GAME 02 - EARTH UNDER WATCH

Genre: Reaction / visual anomaly detection

Tagline: Something changed. Can you spot it?

Target duration: 45-75 seconds

Owner: You

CORE LOOP: WATCH -> SPOT -> CLICK -> COMBO -> ESCALATE

One-line pitch

The player watches a satellite/environmental monitoring feed. A subtle change appears. The player must click the anomalous region as quickly as possible. Consecutive correct detections build a combo and increase score pressure.

EARTH UNDER WATCH / LIVE FEED

Satellite feed area

ANOMALY SCAN status

Timer / streak

Round structure

- Preload one base image and one altered state, never wait for image downloads between rounds.
- Show a short READY cue (500-800 ms).
- Show the base image briefly or continuously depending on difficulty.
- Inject the anomaly using a precomputed mask, image swap, overlay, or local pixel transform.
- Player clicks the target region before the timer expires.
- Success: freeze target area, show points, increment combo, immediately move to the next round.
- Miss: break combo, apply small penalty, continue. Do not force a long failure animation.

Anomaly families

- Vegetation loss or recovery
- Flooded area expansion
- New burn scar

- Urban footprint change
- Snow/ice boundary change
- Water extent change
- Thermal-style anomaly visualization
- Cloud / atmospheric feature movement

Scoring

Base hit: 500. Speed bonus: up to +300. Combo: x1.0, x1.25, x1.5, x2.0, then capped. False click: -100 or short time penalty. Final round: use a higher difficulty modifier rather than simply a longer timer.

Difficulty ladder

- Easy: large anomaly, long click window.
- Medium: smaller anomaly, moderate visual noise.
- Hard: decoy regions and two similar textures.
- Expert: rapid zoom/pan, smaller target, shorter detection window.

Critical gameplay rule

The game is about visual attention and change detection, not about asking the player to know the correct disaster name. The player should win because they noticed the anomaly, not because they memorized terminology.

06 GAME 03 - SPECTRUM SHIFT

Genre: Interactive spectral / image-compositing puzzle

Tagline: The Earth has more colors than your eyes can see.

Target duration: 45-75 seconds

Owner: Teammate 2

CORE LOOP: TARGET -> EXPERIMENT -> COMPOSE -> REVEAL -> LOCK IN

One-line pitch

The player is given a target such as vegetation, water, burn scar, moisture, or heat anomaly. They manipulate available bands or channels until the target becomes visually distinctive. The game rewards discovery rather than memorization.

SPECTRUM SHIFT / COMPOSITOR

Band controls: R / G / B / NIR / SWIR / Thermal

Live composite image

Target prompt

Core interaction

- Start with a base scene and a target objective.
- Provide 3-5 available source bands depending on level.
- Let the player assign bands to display channels using click, drag, or tap.
- Update the preview instantly; keep the interaction responsive and low latency.
- Optionally show a small target visibility meter that rises when the target becomes more separable.
- Player presses LOCK IN when they believe the target is sufficiently visible.
- Award score based on closeness to an intended band combination, time used, and number of unnecessary changes.

Important design decision

Do not begin by teaching the mapping. The game should let the player discover that different wavelength information changes what stands out. The explanatory copy appears after success, for 1-2 seconds, as the takeaway.

Targets

- Vegetation
- Water
- Burn scar
- Urban area
- Snow / ice
- Surface moisture
- Thermal hotspot

Level design

- Level 1: 3 channels, one obvious target.
- Level 2: 4 channels, multiple plausible outputs.
- Level 3: 5+ channels, time limit.
- Level 4: two targets, choose which one to optimize.
- Optional challenge: unknown band label where the player infers its utility from behavior.

Scoring

Target quality: 0-700. Speed: 0-200. Efficiency: 0-100. The intended combination should not be the only way to win; reward any valid composite that makes the target sufficiently distinct. This prevents the game from becoming a memorization test.

Success criteria

A first-time player should experience an obvious visual "aha" moment within the first level. No more than two sentences of instruction should be needed before play.

07 Content, Imagery & Asset Preparation

The quality of the games is strongly dependent on the quality of the visual dataset. Content should be prepared before UI coding is considered complete. Use public-domain, government, or appropriately licensed datasets and retain attribution metadata for every asset.

Asset type	Recommended quantity	Requirements
Pixel Hunter scenes	40-60 images	Mix of land, water, cryosphere, atmosphere, disasters; difficulty labels.
Before/after pairs	15-20 pairs	Strong visible changes; consistent crop and alignment.
Earth Under Watch bases	20-30 bases	Precomputed anomaly masks; 2-3 difficulty variants each.
Spectrum scenes	10-15 scenes	Multiband imagery with validated channel order and metadata.
UI assets	Small shared set	Logo lockup, station badges, icons, success/fail indicators.

Recommended content metadata

id, title, theme, source, license, crop, difficulty, correctAnswer, distractors, timeLimit, revealBudget, targetMask, intendedBands, explanation. Keep all content in structured JSON so the team can expand the game after launch without touching gameplay logic.

Image processing requirements

- Normalize aspect ratio and crop once during preprocessing.
- Generate thumbnails or compressed web versions for fast loading.
- Keep source files outside the runtime bundle.
- For anomaly games, precompute masks instead of doing expensive full-image diffs on every frame.
- For spectral games, store channels as aligned arrays/images with identical dimensions.

08 UI / UX Specification

The visual system should feel modern, technical, and energetic without becoming a science-fiction parody. Use dark navy foundations, bright accent colors per game, large typography, thin grid lines, and restrained motion.

Element	Rule
Navigation	Only START, HELP, REPLAY, NEXT; avoid nested menus during play.
Timer	Always visible; use numeric time plus a subtle progress indicator.
Primary action	One visually dominant action per state.
Feedback	Every correct/incorrect action produces immediate animation + score delta.
Accessibility	High contrast, large click targets, do not rely on color alone for correctness, avoid rapid flashing.
Audio	Optional; short click, success, miss, combo, and reveal sounds. Provide mute.
Error state	Never expose technical errors to players; silently retry assets or switch to fallback content.

Example result card

RESULT / SCORE

FINAL SCORE

BEST COMBO

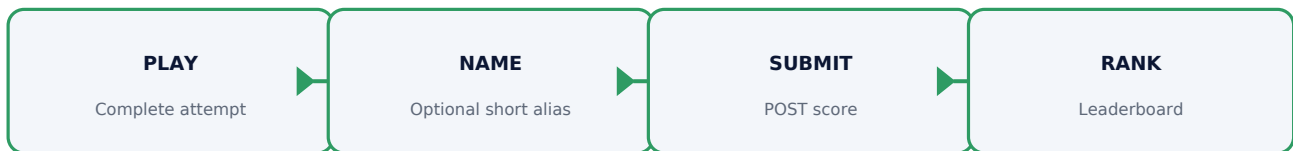
ONE-LINE GRSS TAKEAWAY

PLAY AGAIN / NEXT GAME

09 Shared Scoring & Leaderboard

The leaderboard is a shared event layer, not a shared gameplay dependency. Each game can finish and continue even if the network is temporarily unavailable.

Player flow



Suggested score schema

playerId, displayName, gameId, score, durationMs, difficulty, timestamp, sessionId, signature. At minimum, store name, game, score, duration, and timestamp. Use server-side validation for score ranges to reduce accidental or malicious leaderboard pollution.

Leaderboard behavior

- Show top 10 on the booth display.
- Allow a player to enter a short nickname after the result, not before gameplay.
- Do not require email, phone number, or account creation.
- Normalize score formats across games so that one game does not dominate purely because its numeric score scale is larger.
- Optionally publish a "Top 3 per game" board and an overall board using normalized score percentiles.

Offline fallback

Each station maintains a local high-score cache. When connectivity returns, scores can sync. If the network is fully unavailable, the volunteer can still run the games and show the local station leaderboard.

10 Team Development Plan

The cleanest structure is three parallel workstreams with a shared contract established on day one.

Workstream	Primary owner	Core deliverables	Dependencies
Earth Under Watch	You	Gameplay engine, anomaly injection, combo scoring, difficulty scaling, polish.	Shared UI + score API
Pixel Hunter	Teammate 1	Tile reveal system, image content pipeline, scoring, time-shift mode.	Shared UI tokens
Spectrum Shift	Teammate 2	Band compositor, target scoring, channel UI, spectral content pack.	Shared image metadata
Shared infrastructure	All / one designated integrator	Leaderboard API, data schema, branding, deployment, analytics.	All three games

Recommended implementation sequence

- Day 1: shared design tokens, routes, score contract, content schema, screen dimensions.
- Day 2-3: playable vertical slice for each game, one representative content item each.
- Day 4-5: content expansion, difficulty logic, feedback animation, mobile/touch checks.
- Day 6: shared leaderboard, offline fallback, score normalization.
- Day 7: booth simulation with timed players, observe queues, fix friction.

Build a playable vertical slice first. Do not spend a week perfecting UI around an untested game loop.

11 Testing & Acceptance Criteria

Testing must be done with real students or teammates who have not seen the rules. The key metric is not code correctness alone; it is whether the game works when the player is distracted, rushed, and unfamiliar with GRSS.

Test	Pass condition
Onboarding	Player can begin within 10 seconds of reaching the station.
Instruction	Volunteer needs no more than one sentence to explain the mechanic.
Latency	Clicks, reveals, image swaps, and channel changes respond visually in under ~100 ms on booth hardware.
Throughput	Average complete attempt stays within target duration; result screen does not create a queue bottleneck.
Replay	Content rotation prevents immediate repetition; players can restart with no page reload.
Failure recovery	Missing asset / API outage does not crash the station.
Touch	All primary controls remain usable on touchscreen if that mode is deployed.
Content validity	Every scenario has verified answer, source, license, and explanation metadata.

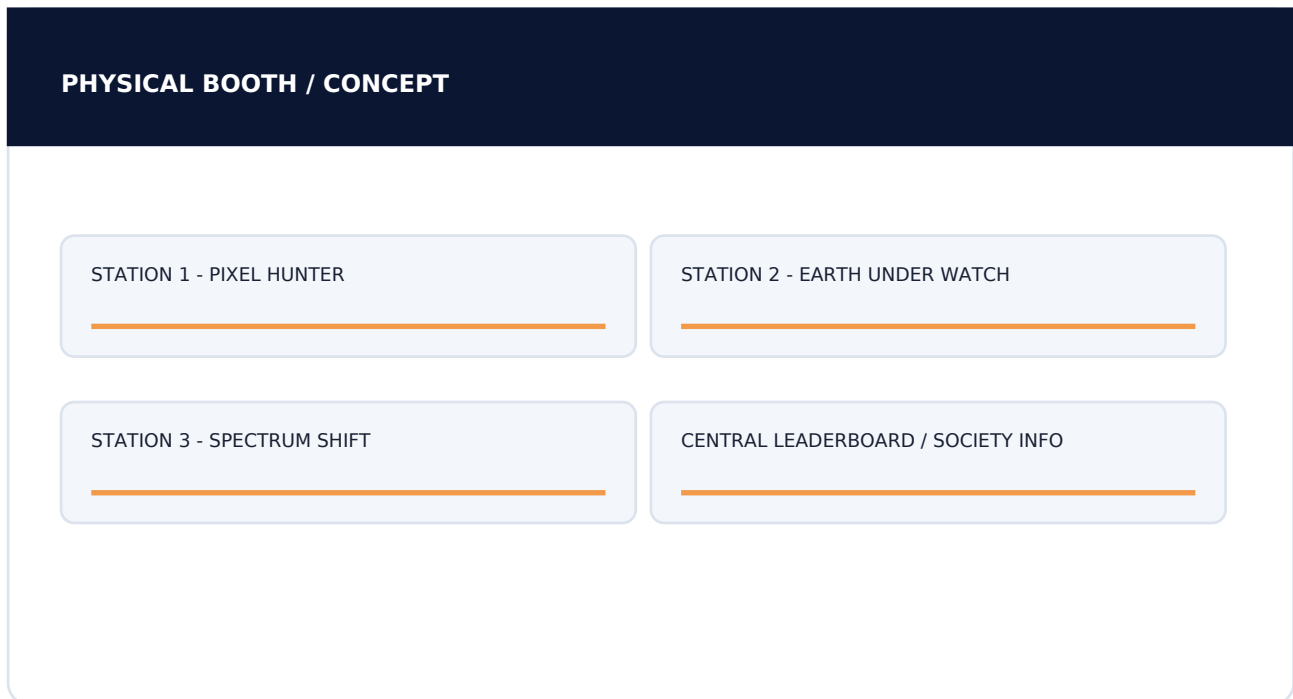
Bug priority for event day

P0: cannot start / crashes / score corrupt. P1: gameplay feels unfair or confusing. P2: visual polish issue. P3: optional feature. Only P0/P1 defects should block release.

12 Open Day Booth Operations

The booth is a live system with people, queues, noise, and interruptions. Design the operating model as carefully as the software.

Recommended station layout



Volunteer script

- Pixel Hunter: "Reveal tiles and guess what the satellite image shows. Fewer reveals means more points."
- Earth Under Watch: "Watch the feed, spot the change, and click it as fast as possible."
- Spectrum Shift: "Change the bands until the target becomes easiest to see, then lock it in."

Queue policy

- One attempt per turn, then rejoin the queue if desired.
- Keep result screens under 5 seconds.
- Display a visible leaderboard to create passive competition while students wait.
- Have one volunteer answer society questions while another manages game flow.

Event-day reliability checklist

- Laptops charged and plugged in where possible.
- Local asset cache tested with internet disconnected.
- Browser fullscreen and notifications disabled.
- Backup mouse(s), charger(s), hotspot or local router if needed.
- Printed QR/URL backup for each game.
- Reset-all button hidden from normal players but accessible to volunteers.

13 Learning Takeaways Without Lecturing

Every game needs a one-line explanation that appears after successful play. The line should answer: "What did I just experience that is actually GRSS?"

Game	Player experience	Post-play takeaway
Pixel Hunter	Recognize an Earth feature from partial imagery.	Remote sensing lets us interpret large areas of Earth from data collected without direct physical contact.
Earth Under Watch	Spot a small change among large amounts of visual information.	Remote sensing can support monitoring of environmental change and natural hazards over wide areas.
Spectrum Shift	Experiment with different spectral channels to reveal a target.	Different parts of the electromagnetic spectrum reveal information that ordinary visible-light images can miss.

Positioning principle

The game should create curiosity first and explain second. The educational message is the reward for playing, not a prerequisite for starting.

14 Feature Roadmap

Priority	Feature	Why
P0	Core gameplay + score	Without this there is no game.
P0	Content rotation	Prevents immediate repetition.
P0	Error recovery / offline mode	Protects the booth from network failures.
P1	Leaderboard	Creates repeat-play motivation.
P1	Combo / streak feedback	Improves game feel.
P1	Difficulty scaling	Keeps mixed audiences challenged.
P2	Sound effects	Adds polish but should not block launch.
P2	Advanced analytics	Useful for learning what content performs best.
P3	Post-event online mode	Turns booth games into reusable society assets.

Post-event potential

These games can later become evergreen outreach tools: a GRSS chapter website mini-game page, school outreach demo, orientation activity, recruitment campaign, or social-media challenge. The content layer should therefore be designed for expansion from the beginning.

15 GRSS Reference Grounding

The following official GRSS pages were used to anchor the educational scope represented in this specification.

GRSS - About

Society mission, member profile, interdisciplinary character, fields of interest.

<https://www.grss-ieee.org/about/>

What is Geoscience and Remote Sensing?

Definition of remote sensing and examples including wildfire mapping, cloud/weather monitoring, urban and land-cover change.

<https://www.grss-ieee.org/about/what-is-geoscience-and-remote-sensing/>

IEEE GRSS Education Portfolio

Student engagement, hands-on training, Student Grand Challenges, and outreach.

<https://www.grss-ieee.org/resources/education/>

IEEE GRSS 2026 President Message

2026 theme: "Amplify the Insight - Elevate the Impact."

<https://www.grss-ieee.org/about/message-from-our-president/>

Scope of GRSL

Detailed scope covering Earth systems, remote sensing, processing, interpretation, and dissemination.

<https://www.grss-ieee.org/publications/author-resources/scope-of-grsl/>

FINAL TEAM DECISION: build three genuinely different games - Pixel Hunter, Earth Under Watch, and Spectrum Shift - with a shared visual system and lightweight leaderboard.